

Declaring multiple record-type files

Click to view.

When viewing , left click (or press the Page Up key) to see the next frame and right click (or press Page Down) to see the previous frame.

InsertRec

StudentId	StudentName	DateOfBirth	CourseCode	Gender
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9	8	2	5	3	2	7	C	O	U	G	H	L	A	N	M	S	1	9	5	2	0	9	1	0	L	M	5	1	M
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

When a file contains multiple record types, a record declaration (starting with a 01 level number) must be created for each type of record.

```
FD TransFile.
01 InsertRec.
  02 StudentId          PIC 9(7) .
  02 StudentName.
    03 Surname          PIC X(8) .
    03 Initials         PIC XX.
  02 DateOfBirth.
    03 YOBirth          PIC 9(4) .
    03 MOBirth          PIC 99.
    03 DOBirth          PIC 99.
  02 CourseCode         PIC X(4) .
  02 Gender             PIC X.
```

InsertRec

StudentId	StudentName	DateOfBirth	CourseCode	Gender
-----------	-------------	-------------	------------	--------

9	8	2	5	3	2	7	C	O	U	G	H	L	A	N	M	S	1	9	5	2	0	9	1	0	L	M	5	1	M
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

StudentId

DeleteRec

Even though there are different types of record in the file, and separate record declarations have been created for each record type, only a single 'record buffer' is created for the file.

```
FD TransFile.
01 InsertRec.
02 StudentId          PIC 9(7) .
02 StudentName.
03 Surname            PIC X(8) .
03 Initials           PIC XX.
02 DateOfBirth.
03 YOBirth            PIC 9(4) .
03 MObirth            PIC 99.
03 DOBirth            PIC 99.
02 CourseCode         PIC X(4) .
02 Gender             PIC X.

01 DeleteRec.
02 StudentId          PIC 9(7) .
```

InsertRec

StudentId	StudentName	DateOfBirth	CourseCode	Gender
-----------	-------------	-------------	------------	--------

9	8	2	5	3	2	7	C	O	U	G	H	L	A	N	M	S	1	9	5	2	0	9	1	0	L	M	5	1	M
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

StudentId
DeleteRec

StudentId	NewCourseCode
-----------	---------------

UpdateRec

All the record declarations map to this area of storage, which is the size of the largest record, and all the identifiers are current/active at the same time.

```
FD TransFile.
01 InsertRec.
02 StudentId          PIC 9(7) .
02 StudentName.
03 Surname            PIC X(8) .
03 Initials           PIC XX.
02 DateOfBirth.
03 YOBirth            PIC 9(4) .
03 MObirth            PIC 99.
03 DOBirth            PIC 99.
02 CourseCode         PIC X(4) .
02 Gender             PIC X.

01 DeleteRec.
02 StudentId          PIC 9(7) .

01 UpdateRec.
02 StudentId          PIC 9(7) .
02 NewCourseCode      PIC X(4) .
```

InsertRec

StudentId	StudentName	DateOfBirth	CourseCode	Gender
-----------	-------------	-------------	------------	--------

9	8	4	1	3	2	5	L	M	6	2	H	L	A	N	M	S	1	9	5	2	0	9	1	0	L	M	5	1	M
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

StudentId
DeleteRec

StudentId	NewCourseCode
-----------	---------------

UpdateRec

When a record smaller than the size of the largest record is read into the buffer, any data that is not explicitly overwritten is left intact.

```
FD TransFile.
01 InsertRec.
02 StudentId          PIC 9(7) .
02 StudentName.
03 Surname            PIC X(8) .
03 Initials           PIC XX.
02 DateOfBirth.
03 YOBirth            PIC 9(4) .
03 MOBirth            PIC 99.
03 DOBirth            PIC 99.
02 CourseCode         PIC X(4) .
02 Gender             PIC X.

01 DeleteRec.
02 StudentId          PIC 9(7) .

01 UpdateRec.
02 StudentId          PIC 9(7) .
02 NewCourseCode      PIC X(4) .
```

InsertRec

StudentId	StudentName	DateOfBirth	CourseCode	Gender
-----------	-------------	-------------	------------	--------

9	6	2	7	3	9	2	L	M	6	2	H	L	A	N	M	S	1	9	5	2	0	9	1	0	L	M	5	1	M
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

StudentId
DeleteRec

StudentId	NewCourseCode
-----------	---------------

UpdateRec

When a record smaller than the size of the largest record is read into the buffer, any data that is not explicitly overwritten is left intact.

```
FD TransFile.
01 InsertRec.
02 StudentId          PIC 9(7) .
02 StudentName.
03 Surname            PIC X(8) .
03 Initials           PIC XX.
02 DateOfBirth.
03 YOBirth            PIC 9(4) .
03 MObirth            PIC 99.
03 DOBirth            PIC 99.
02 CourseCode         PIC X(4) .
02 Gender             PIC X.

01 DeleteRec.
02 StudentId          PIC 9(7) .

01 UpdateRec.
02 StudentId          PIC 9(7) .
02 NewCourseCode      PIC X(4) .
```